

MD RAHIDUL ISLAM

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🌐 LinkedIn | 🐙 GitHub | 📁 Portfolio

About Me

A Computer Science and Engineering student and aspiring full-stack developer, hardworking, detail-oriented, and passionate about technology, with a strong team-player mindset built through academic and team projects. Hands-on experience building web applications using Next.js, NestJS, TypeScript, and PostgreSQL — including responsive interfaces, REST APIs, authentication, and role-based access control — consistently delivered by working closely within collaborative teams to meet deadlines. I value clean, maintainable code and practical problem-solving, and I'm seeking an entry-level software development role where I can contribute to production-quality products while growing alongside an experienced engineering team.

Projects

ChargeHubEV | *Next.js, NestJS, TypeScript, PostgreSQL, Tailwind CSS* | 🐙 [Backend](#) | 🐙 [Frontend](#)

- Developed a full-stack EV charging platform with a responsive Next.js interface and modular NestJS backend.
- Built REST APIs for users, bookings, payments, authentication, and technician workflows using TypeORM and PostgreSQL.
- Implemented JWT authentication, bcrypt hashing, role-based access control, validation, and reusable frontend forms.

Smart Metro E-Ticket Management System | *PHP, JavaScript, AJAX, MySQL, HTML, CSS* | 🐙 [Source Code](#)

- Developed a metro ticketing platform with account management, fare calculation, booking, and ticket validation.
- Used AJAX and MySQL for dynamic, database-driven operations without unnecessary page reloads.

Java Tour Management System | *Java, Object-Oriented Programming* | 🐙 [Source Code](#)

- Created a tour booking application for managing packages, customers, reservations, and payments.
- Applied encapsulation, inheritance, abstraction, and modular class design for maintainable code.

Smart Coffee Shop Management System | *C#, .NET Framework, MySQL*

- Built a desktop system for order processing, billing, inventory tracking, and customer records.
- Designed the relational database and connected it to a modular C# interface for real-time data operations.

Additional Projects

Smart Railway Station Visualization | *C++, OpenGL, GLUT* | 🐙 [Source Code](#)

Course: Computer Graphics, AIUB

- Created an animated railway-station simulation with trains, tracks, platforms, and environmental elements.
- Applied geometric transformations, event handling, rendering, and timer-based animation using OpenGL and GLUT.

Smart Health Monitoring System | *ESP32, Sensors, IoT, Blynk*

Course: Microprocessor and Embedded Systems, AIUB

- Built an ESP32-based system to collect sensor readings and transmit live health data wirelessly.
- Integrated embedded hardware with an IoT dashboard for remote monitoring and device-status updates.

Skills

Computer & Technical Skills:

- **Programming Languages:** Python, JavaScript, TypeScript, C++, C#, Java, PHP, HTML, CSS.
- **Frameworks & Libraries:** Node.js, NestJS, Next.js, React.js, TypeORM, Material-UI, DaisyUI, Tailwind CSS, .NET, GUNA UI, Swing, OpenGL.
- **Databases & Tools:** PostgreSQL, MySQL, Oracle SQL, Git, GitHub, Postman, VS Code, Visual Studio, Linux (Ubuntu/WSL), Blynk, Figma, Trello, Canva, Notion, Cisco Packet Tracer, Multisim, AutoCAD, Microsoft 365.
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networking, Operating Systems, REST API Development, Authentication and Authorization.

Language Skills:

- Bangla: Native
- English: Professional working proficiency
- Hindi: Conversational

Research Experience

Industrial Defect Localization with Explainable Heatmaps

 [Repository](#)

| Focus: Computer Vision, Explainable AI | Tech: Python, PyTorch, TensorFlow, OpenCV

- Compared Grad-CAM and patch-based heatmaps for lightweight, explainable defect localization on MVTec AD.
- Built an image-preprocessing and CNN evaluation pipeline and applied explainability techniques to highlight defective regions.

Adversarially Robust Network Intrusion Detection

 [Repository](#)

| Focus: Machine Learning, Cybersecurity | Tech: Python, Random Forest, XGBoost, Neural Networks

- Designed a multi-agent intrusion-detection concept on NSL-KDD using ensemble and neural network models.
- Performed data preprocessing, threshold tuning, and model evaluation for a robust detection pipeline.

Educational Background

Bachelor of Science (BSc), American International University-Bangladesh

Department: Computer Science and Engineering

Dhaka, Bangladesh

Expected: December 2026

Higher Secondary Certificate (HSC), Notre Dame College, Mymensingh

Department: Science

Mymensingh, Bangladesh

2021

Secondary School Certificate (SSC), Poncahashi High School

Department: Science

Mymensingh, Bangladesh

2019

Extracurricular Activities

General Member, Notre Dame Science Club

2019–2020

- Assisted with the National Science Festival and participated in workshops, exhibitions, and academic seminars.

Parade Team Leader, Poncahashi High School

2017–2018

- Led practice sessions, coordinated members, and maintained discipline during national and school events.

Achievements

Two-time 1st Prize, Inter-School Parade Competition

2017–2018

Led the Poncahashi High School parade team to first place twice.

1st Place, School-Level Scholarship Examination

2017–2018

Hobbies

- Traveling, Reading books, Playing guitar, Watching movies